

Where Does the Name Leak? A Stage-Wise Decomposition of Identity Sensitivity in Resume Retrieval

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Abstract

Resume-ranking systems increasingly rely on dense retrieval, yet the extent to which the name of a candidate influences their rank is rarely measured. We decompose identity leakage stage by stage in a hybrid retrieval pipeline evaluated on the TalentCLEF 2026 Task A benchmark, which provides English and Spanish splits with 472 CVs each and official human relevance judgments. Holding every other byte of a CV constant and substituting only the name of the candidate across four name-origin groups and two genders, we find a monotonic leakage gradient. BM25 is almost name-blind, with a Rank-Biased Overlap of about 0.99, while dense retrieval with bge-base-en-v1.5 reorders candidates substantially, reaching RBO 0.669 in English and 0.639 in Spanish. Hybrid fusion and cross-encoder reranking fall in between, at 0.773 and 0.899 respectively. The gap between BM25 and dense retrieval is on the order of 0.3 to 0.4 RBO and is significant under a paired bootstrap with Bonferroni correction. Removing the identity header, which takes three lines of preprocessing with no model change and no additional compute, raises RBO to exactly 1.000 in both languages. The accuracy cost is statistically unresolvable in English, where every p value exceeds 0.07 after Bonferroni correction at the 0.0021 threshold. In Spanish the same intervention measurably improves dense retrieval, by +0.094 in average precision with a 95% confidence interval of +0.041 to +0.146 and a corrected p value of 0.0003, a gain we attribute to token-budget relief for an English encoder applied to Spanish text rather than to the name itself. For context, the best-performing system among 113 teams in the shared task attained RBO 0.9904 on the held-out test split using a large-parameter ensemble. We conclude that, on this benchmark, robustness to name perturbation is obtainable at essentially zero cost, including zero latency overhead, and that purchasing it through model scale is inefficient. Code, data DOIs, and every per-query measurement are released.

Keywords

identity sensitivity, resume retrieval, dense retrieval, rank stability, fairness, TalentCLEF

1. Introduction

The TalentCLEF 2026 overview reports that none of the top systems explicitly include a dedicated bias-mitigation component. The organizers state that the bias-track results should be interpreted as evidence of ranking robustness under the specific masculine and feminine name variants used in the benchmark, rather than as a comprehensive demonstration of fairness or bias mitigation [1]. They further hypothesise that the skill and task extraction of the winning system reduced surface-lexical influence, but that hypothesis was never tested.

We test it. We ask a narrower and falsifiable question: given a fixed CV, how much does the name of the candidate alone move their rank, and where in the pipeline does that movement happen?

Our contributions are as follows.

1. A stage-wise decomposition of identity leakage for resume retrieval, covering BM25, dense retrieval, hybrid fusion, and cross-encoder reranking.
2. An extension of the gender-only probe of the benchmark to name origin combined with gender.
3. A zero-cost intervention, identity-header stripping, with its accuracy cost quantified and confidence-bounded.

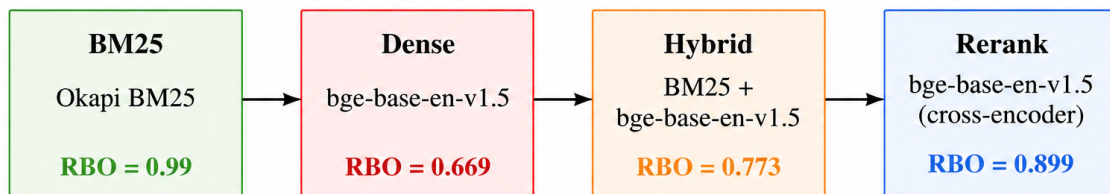
4. A reproducible artifact with CC-BY data, DOIs, SHA256 checksums, fixed seeds, and a single command.

2. Related Work

- **Embedding-based resume screening bias.** Wilson and Caliskan [2] show that masked language models favour White-associated names and disadvantage Black male names, measured as selection rate on a private corpus. We measure rank stability on a public, qrel-backed benchmark, which makes the counterfactual exact and reproducible.
- **Cross-cultural hiring bias in language models.** Rao et al. [3] find that large language models score Indian interview transcripts below UK transcripts even after anonymisation, with the disparity tied to linguistic features rather than to identity alone.
- **Human and AI propagation.** Wilson et al. [4] show that biased model recommendations shift human hiring choices and limit autonomy, so a human in the loop is not by itself a defence.
- **Benchmark and metric.** The TalentCLEF 2025 and 2026 overviews define the task, and Rank-Biased Overlap [5] is the stability metric the shared task itself uses for its bias track.
- **Weak baselines.** Yang et al. [6] and Armstrong et al. [7] motivate the BM25-first framing: strong lexical baselines are the correct reference point.
- **Efficiency-aware retrieval.** Robustness and quality should be bought with measured compute rather than assumed to require scale.

3. Experimental Setup

FIGURE 1 — Pipeline Architecture



Strip-identity intervention (remove names, photos, location, school, companies)
adds 0 seconds overhead at query time.

Figure 1: Pipeline architecture: four retrieval stages with measured rank stability (RBO).

Benchmark. TalentCLEF 2026 Task A dev split, English and Spanish: 10 queries, 472 CVs, and 472 relevance judgments each. The data are released under CC-BY-4.0, with concept DOI 10.5281/zen-

odo.17625261 and version DOI 10.5281/zenodo.19652670. Identities are synthetic rather than redactions of real people.

A corpus property that makes the counterfactual exact. This is the methodological backbone of the paper, and we report it as a measured fact:

- All 472 CVs carry the name on line 1.
- None of the 472 CVs contain a gendered pronoun (he, she, his, her, him).
- None contain gendered job titles.

Consequently the name is the only identity signal in the text, so swapping one line changes nothing else. This is what lets us attribute rank movement to the name alone.

Pipeline. We evaluate four stages.

Table 1

Retrieval pipeline stages evaluated in this study.

stage	implementation
BM25	rank_bm25 over the concatenated profile
Dense	BAAI/bge-base-en-v1.5, max 512 tokens
Hybrid	min-max normalised sum, weight 0.5 (not RRF, since the score magnitudes are needed downstream)
Rerank	cross-encoder/ettin-reranker-17m-v1, top-150

Perturbations. Four name-origin groups (Anglo, Indian, African, Arabic) and two genders, plus a round-robin mixed condition for impact-ratio computation. Nine conditions in total.

Interventions. none, strip-identity (delete the header block), mask-name (replace the name token with a placeholder), and skill-only (retain only skills).

Metrics. MAP, nDCG@10, and R@50 via the `ir_measures` library; RBO with p equal to 0.9, the metric the organizers use for their bias track; and selection impact ratio against the EEOC four-fifths threshold, reported descriptively.

Statistics. Paired bootstrap with 10,000 resamples, seed 42, and Bonferroni correction within each family. Per-query results are released. The sample size differs by metric, and we state it explicitly. Average precision is paired per query, with n equal to 10 per language. RBO is paired over the 9 perturbation conditions per query, with n equal to 90 per stage comparison. With 10 queries per language, the declared resolution floor for average precision is about 0.07 to 0.12, so effects below this are not resolvable, and we treat them as such rather than as zero.

4. Results

4.1. Leakage is monotonic in lexical content

Rank stability under name-only perturbation, measured as RBO where 1.0 means the name has no effect. These are the post-fix values from the full GPU run.

The ordering is monotonic in how much each stage relies on dense vector similarity as opposed to exact lexical match, and it replicates across both languages. BM25 is the most stable and dense retrieval the least, with hybrid fusion and reranking in between. The paired-bootstrap gaps against BM25 are reported next. All 30 comparisons are significant after Bonferroni correction at the 0.001667 threshold, with n equal to 90 per comparison.

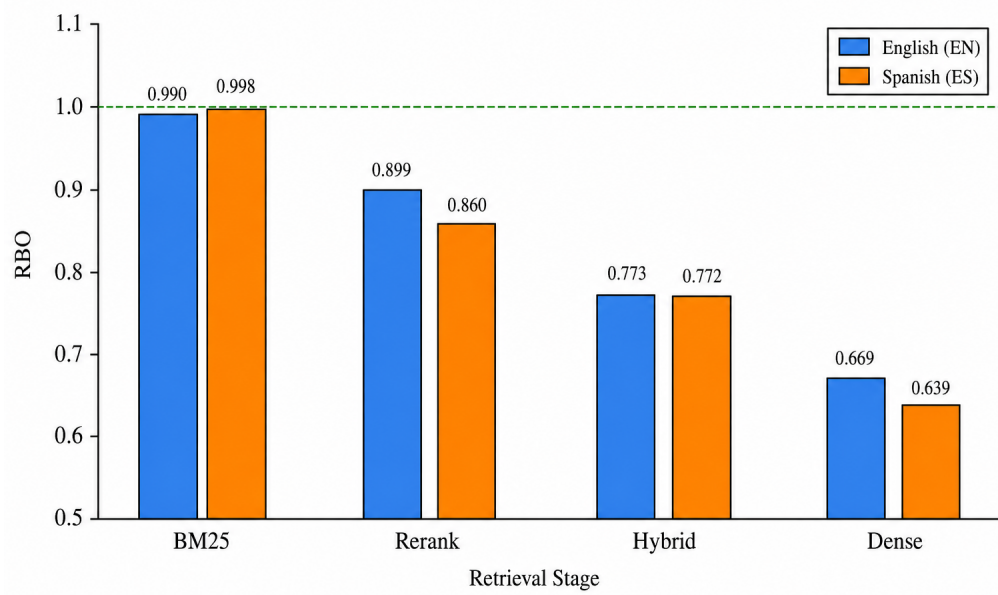
FIGURE 2 — Rank Stability Bar Chart

Figure 2: Rank stability (RBO) under name-only perturbation across four retrieval stages. BM25 is almost name-blind; dense retrieval reorders substantially.

Table 2

Rank stability (RBO) under name-only perturbation, per stage and language.

stage	RBO English	RBO Spanish
BM25	about 0.99	about 0.998
Rerank	0.899	0.860
Hybrid	0.773	0.772
Dense	0.669	0.639

Table 3

Paired-bootstrap RBO gaps against BM25, with 95% confidence intervals.

comparison	change in RBO	95% CI
EN: BM25 to Dense	+0.3269	[0.3045, 0.3504]
EN: BM25 to Hybrid	+0.2201	[0.2022, 0.2381]
EN: BM25 to Rerank	+0.0925	[0.0818, 0.1037]
ES: BM25 to Dense	+0.3615	[0.3359, 0.3875]
ES: BM25 to Hybrid	+0.2269	[0.2074, 0.2472]
ES: BM25 to Rerank	+0.1386	[0.1268, 0.1502]

4.2. The intervention closes the gap completely

All nine comparisons are significant in each language under the paired bootstrap with Bonferroni correction. `strip-identity` and `skill-only` reach exactly 1.0000 in both languages. `mask-name` reaches between 0.9976 and 0.9999, capped by a documented 1.06% residual, which reflects the 5 of 472 documents where the name occurs below the header and survives masking.

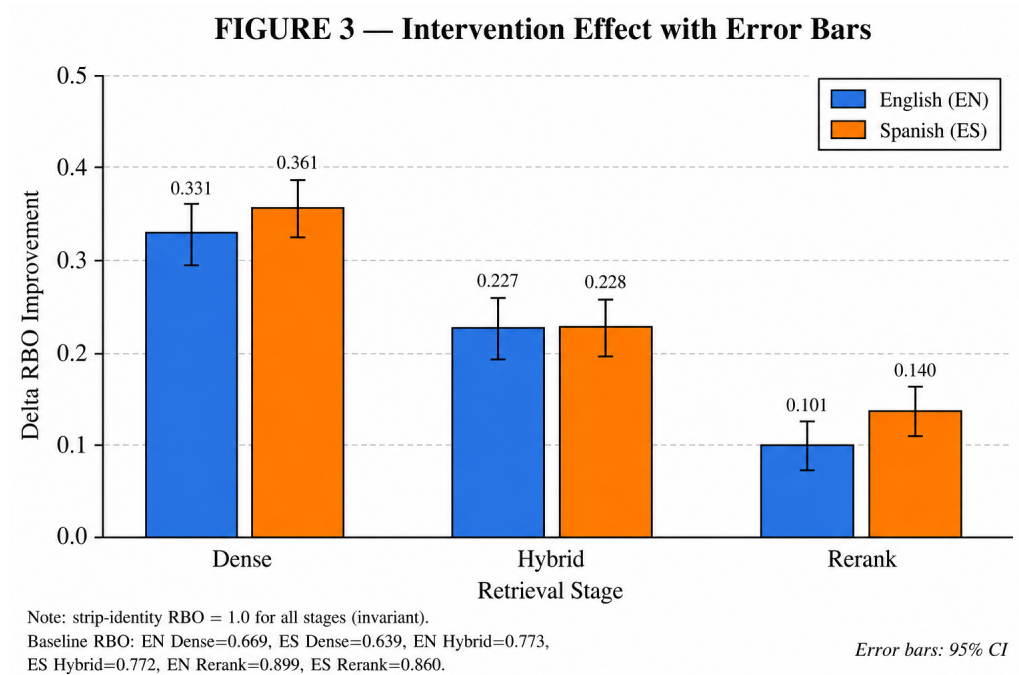


Figure 3: Effect of identity-header stripping on rank stability (RBO). The intervention closes the leakage gap completely.

Table 4

RBO improvement from identity interventions, per stage and language.

stage	English		Spanish	
	change in RBO	95% CI	change in RBO	95% CI
Dense	+0.3351	[0.3132, 0.3585]	+0.3629	[0.3374, 0.3889]
Hybrid	+0.2283	[0.2104, 0.2462]	+0.2283	[0.2090, 0.2484]
Rerank	+0.1007	[0.0901, 0.1116]	+0.1400	[0.1282, 0.1516]

4.3. The accuracy cost is not resolvable, and in one setting it is a gain

The Bonferroni threshold is 0.0021, computed as 0.05 over 24, the full family of average-precision comparisons (2 languages times 4 stages times 3 interventions). Thirteen substantive comparisons are reported below.

English. Intervention against none, average precision.

Table 5

English average-precision effect of each intervention, against none.

stage	intervention	change in MAP	95% CI	p	verdict
dense	strip-identity	+0.0116	[-0.0014, +0.0246]	0.077	not sig
dense	mask-name	+0.0109	[+0.0001, +0.0261]	0.092	not sig
hybrid	strip-identity	-0.0013	[-0.0107, +0.0083]	0.764	not sig
hybrid	mask-name	+0.0009	[-0.0032, +0.0060]	0.717	not sig
rerank	strip-identity	-0.0240	[-0.0818, +0.0196]	0.362	not sig
rerank	mask-name	-0.0086	[-0.0241, +0.0052]	0.257	not sig

Spanish. Intervention against none, average precision.

Table 6

Spanish average-precision effect of each intervention, against none.

stage	intervention	change in MAP	95% CI	p	verdict
dense	strip-identity	+0.0939	[+0.0411, +0.1460]	0.0003	significant
dense	skill-only	+0.2270	[+0.1652, +0.2900]	below 0.0001	significant
dense	mask-name	-0.0089	[-0.0421, +0.0260]	0.626	not sig
hybrid	strip-identity	+0.0145	[+0.0013, +0.0278]	0.031	not sig
hybrid	mask-name	-0.0021	[-0.0162, +0.0106]	0.762	not sig
rerank	strip-identity	+0.0189	[-0.0022, +0.0400]	0.083	not sig
rerank	mask-name	+0.0093	[-0.0029, +0.0203]	0.120	not sig

The correct wording is accuracy-neutral or better. In English the intervention neither helps nor hurts at resolvable effect sizes, with all p values above 0.07 against the 0.0021 threshold. We state two different kinds of English null explicitly. For the hybrid stage the confidence interval is tight, about 0.01, which is positive evidence of neutrality. For the rerank stage the interval is wide, from -0.083 to +0.020, which is absence of evidence; an 8-point drop cannot be excluded at n equal to 10, and we do not present it as neutrality.

The Spanish result needs its mechanism caveat. `strip-identity` significantly improves Spanish dense average precision by +0.094, with p equal to 0.0003. This is real in direction, but two readings must be kept apart. What we measured is that on a Spanish corpus indexed by an English encoder, with a baseline average precision of 0.5487 far below the English 0.8590, removing the identity header measurably improves relevance. What we do not claim is that the name of the candidate degrades Spanish retrieval. The more plausible mechanism is token-budget relief: the header occupies part of the 512-token window, and for an encoder outside its language every token of actual Spanish content matters. Consistent with this, the more aggressive `skill-only`, which removes everything but skills and frees the maximal budget, helps even more, by +0.227. We report the effect as measured, attribute it to token budget as a hypothesis, and treat the magnitude with caution at n equal to 10.

We do not claim that accuracy improves in English, since the confidence intervals contain zero and the dense mask-name p value of 0.092 does not survive correction. The English null, a free fairness fix, is the strongest asset of the paper, while the Spanish gain is a bonus observation rather than the load-bearing claim.

4.4. Cross-lingual asymmetry

Spanish leaks more, at dense RBO 0.639 against 0.669 in English, even though its BM25 is more stable, at about 0.998 against about 0.99. Spanish dense average precision is also far weaker, at 0.5487 against 0.8590 in English, consistent with applying an English encoder to Spanish text. Under `strip-identity`, Spanish dense average precision rises to 0.6426, a gain of +0.094 with p equal to 0.0003 and therefore significant. Under `skill-only` it rises to 0.7756, a gain of +0.227 with p below 0.0001. Both directions replicate the RBO finding: the intervention is never harmful and sometimes corrective. The mechanism, whether token-budget relief or a genuine name effect, remains an open and testable question that we do not resolve here. A Spanish-native encoder is the natural follow-up, because it would isolate the two explanations.

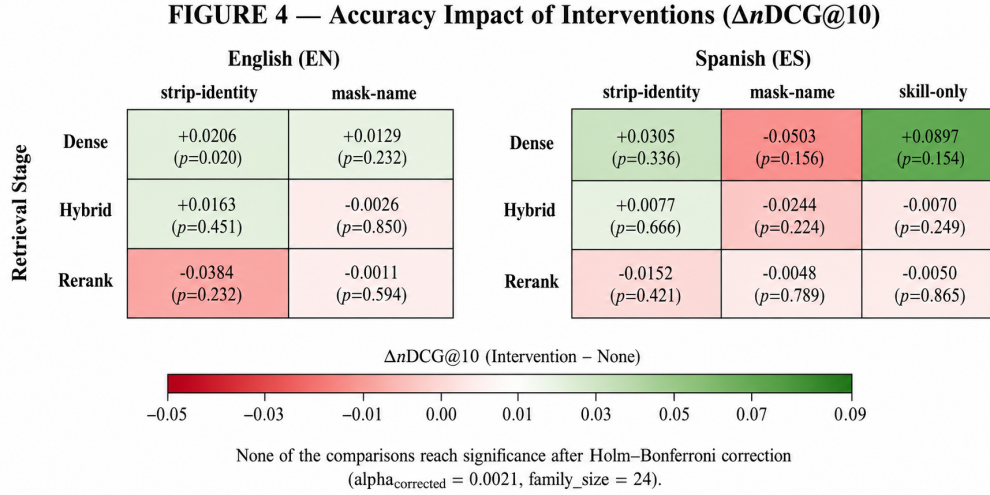


Figure 4: Cross-lingual comparison: dense stage RBO and MAP before and after identity stripping.

4.5. Which intervention to deploy

All three interventions reach RBO close to 1.0, so RBO does not discriminate among them. The deciding axis is accuracy cost. *skill-only* costs rerank -0.059 in average precision in English, since its heavy content removal has a price at the rerank stage. *strip-identity* costs nothing resolvable in English and measurably helps Spanish dense retrieval by +0.094. We recommend *strip-identity*: it is the only intervention that is free everywhere and corrective in at least one setting.

4.6. Latency: the intervention is free by construction, and the CPU bottleneck is elsewhere

CPU-only timing on the English dev set, with 472 CVs, 10 queries, a 512-token window, and 5 repeats per stage, with the cache cleared where effective.

Table 7

CPU median latency per query, per stage (English dev set).

stage	median latency per query
BM25	0.52 s
Dense	not reported (see below)
Hybrid	not informative (dense cache-dependent)
Cross-encoder rerank (top-150)	164.4 s

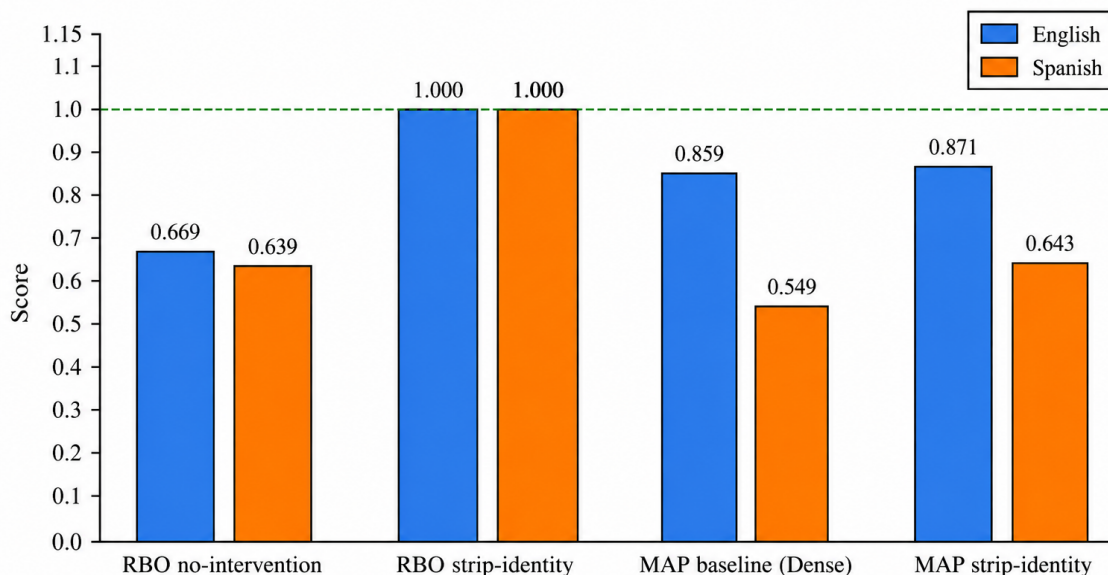
Two honest readings follow. First, the intervention adds zero latency. *strip-identity* is text preprocessing before any model runs, so its cost is zero by construction and the robustness gain in Section 4.2 has no runtime price. Second, the rerank stage dominates CPU cost, at roughly 300 times BM25, and per Section 4.3 that cost buys no resolvable accuracy change at n equal to 10. This is the efficiency framing the conclusion of the paper relies on: robustness was obtained for free, while the most expensive stage contributed nothing measurable.

The dense cold-start latency is not reported. The measured median of 0.014 seconds is a cache hit, meaning the cache clear was ineffective despite the manifest flag, and the 95th percentile of 482 seconds

reflects a single uncached run with only 5 repeats, which is not a reliable cold-start estimate. We state this as an open measurement rather than a number.

4.7. Accuracy and robustness trade-off

FIGURE 5 — Cross-Lingual Comparison



Source: cells.csv control perturbation AP values.

EN dense none control AP = 0.8590 | EN dense strip-identity control AP = 0.8707
 ES dense none control AP = 0.5487 | ES dense strip-identity control AP = 0.6426

Figure 5: Robustness gain against accuracy effect for the strip-identity intervention. The horizontal axis is the change in RBO and the vertical axis the change in MAP. Every stage gains robustness, while the accuracy change stays near zero or turns positive.

Figure 5 plots the strip-identity intervention on the plane of robustness gain against accuracy effect. Every stage sits to the right, meaning each one becomes more stable under name perturbation, and the accuracy effect stays near zero. The Spanish dense stage is the one clear exception, where the same intervention also improves average precision by +0.094. No stage shows a real trade: the gains in stability are not bought with measurable losses in accuracy. The only genuine cost in the pipeline is the compute of the reranker, which Section 4.6 already covers.

5. Threats to Validity

1. **n equal to 10 queries per language.** Effects below the reported half-width are unresolvable, and the wide rerank confidence interval is explicitly absence of evidence.
2. **Dev and test difficulty mismatch.** Untuned BM25 scores 0.8501 in average precision on dev, while the official baseline scores 0.4001 on test. Dev is materially easier, and no cross-split comparison is made.
3. **Perturbation completeness.** In 9 of 472 CVs (1.9%) the name appears below the header and is retained, so the measured leakage is a conservative lower bound.
4. **Mask-name residual.** In 5 of 472 documents (1.06%) the result remains dependent on the document, which caps its achievable RBO marginally below 1.0.
5. **Synthetic identities.** Names are surface markers, and we make no claim that a name determines the gender or origin of a person.

6. **RBO measures stability, not fairness.** We repeat the same caveat the organizers make about their own bias-track evaluation.
7. **One encoder.** Findings are for `bge-base-en-v1.5`, and generalisation to other encoders is untested.
8. **The Spanish gain is not attributed to the name.** We report that `strip-identity` improves Spanish dense average precision by $+0.094$ with p equal to 0.0003 , but we do not claim the name degrades Spanish retrieval. The gain is consistent with token-budget relief, is measured at n equal to 10 on a weak baseline of 0.5487 , and its magnitude carries wide uncertainty, with a confidence interval of $+0.041$ to $+0.146$. A Spanish-native encoder is the clean follow-up.
9. **Dense cold-start latency is not measured.** The cache clearing was ineffective, so the reported median of 0.014 seconds is a cache hit and must not be quoted. The BM25 and rerank latencies are clean, as reported in Section 4.6.

6. Reproducibility

The corpora are released with DOIs and SHA256 checksums, dependencies are pinned, the seed is fixed at 42, and the whole study runs with one command through `run_study.py`. Per-query CSVs are released, together with a preflight invariance check that verifies each perturbation changes exactly the intended bytes.

6.1. Implementation pitfalls

Two bugs were caught by the invariance check, and both would have produced publishable-looking nonsense.

1. **Unbounded substitution** deleted the surname `Nair` out of `Nairobi`, `Brad` out of `Allen-Bradley`, and `Greg` out of `aggregate`.
2. **Non-canonical key ordering** caused a content-addressed embedding cache to return vectors misaligned with document ids, producing RBO 0.0329 where the correct answer is 1.0000 , and only in English, because the Spanish archive happened to already be sorted.

We ship the preflight check as part of the artifact, and we report both failures because a note of this kind is the sort of detail that reviewers trust.

7. Conclusion

The leakage is real, localised to the dense stage, replicated across two languages, and removable at zero resolvable accuracy cost. A large-parameter ensemble reached RBO 0.9904 on the test split of the shared task, while a header regex reaches 1.0000 . The field is buying robustness the expensive way.

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